

34. As per Koppen's classification of climate, which one of the following is the suitable description of North-East India including North Bengal?

- (a) Tropical monsoon rainforest (Amw)
- (b) Sub-tropical monsoon rainforest (Am)
- (c) Tropical wet and dry climate (Monsoon Savannah) (Aw)
- (d) Humid sub-tropical climate with dry winter (Cwg)

Uttarakhand P.C.S. (Pre) 2016

Ans. (d)

As per Koppen's classification of Climate, Humid sub-tropical climate with dry winter (Cwg) is the most suitable description for North-East India including North Bengal, Ganga Plains etc.

35. According to Koppen's climatic classification, the climate of North Bihar may be explained as:

- (a) Cwg
- (b) Aw
- (c) CA'w
- (d) CB'w
- (e) None of the above / More than one of the above

63rd B.P.C.S. (Pre) 2017

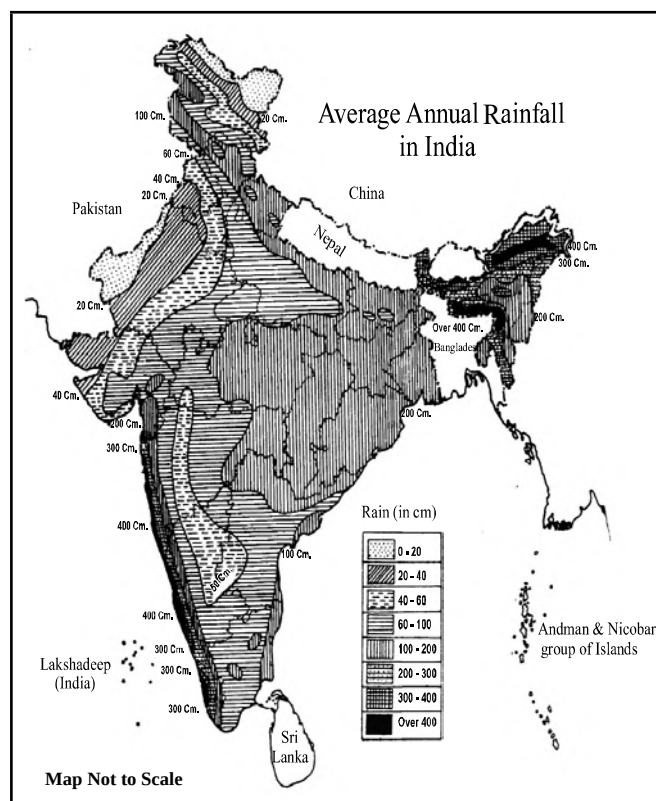
Ans. (a)

According to Koppen's climatic classification, the climate of North Bihar may be explained as Cwg.

ii. Rainfall

*Generally in India rainfall occurs from the month of June to September when south-west monsoon is effective throughout the country. In this period, the low-pressure zone of **North-Western India** is very intense and systematic. Due to low air pressure, south-east winds from the **tropic of Capricorn** in the Southern Hemisphere cross the equator and are attracted towards the Indian Peninsula. Passing over the oceans these winds are filled with humidity and split into two branches - **Arabian Sea branch** and **Bay of Bengal branch**. *When the south-west monsoon winds enter the terrestrial area, it rains with an intense thundering and torrential storms. The arrival of these winds and rainfall is called '**burst**' of the monsoon. *The speed of these winds is more than 30 km/h and they become effective throughout the country within a month. *According to the data Rainfall Statistics of India (2022), almost the entire rainfall is received from the southwest monsoon (June-September). *Monsoon winds of the Arabian Sea are obstructed by the Western Ghats. These winds climb the slope of the Western Ghats and soon become cool causing

heavy rainfall on the windward side of Western Ghats. These winds cause little rainfall to the east of the Western Ghats. The region of low rainfall is known as the **rain-shadow area**.



*Another branch of monsoon winds from the Arabian Sea causes rainfall in the extensive areas of central India. Thereafter, they enter the Ganga plains and mingle with the Bay of Bengal branch. The Bay of Bengal branch of monsoon winds causes rainfall in northeast and Gangetic plains reaching as far as the Punjab plains. *The actual annual rainfall in India is about **1257.0 mm (Data - 2022)** but it has great spatial variations. The highest amount of rainfall occurs along the west coast of the Western Ghats as well as in the Eastern-Himalayan region and Meghalaya. ***Mawsynram** receives the highest rainfall in the world. **Leh** (Ladakh, Union Territory) receives the least rainfall (**6.38cm**) in India. According to the data Rainfall Statistics of India 2022, annually, the District of East Khasi Hills in Meghalaya received the highest rainfall of 8084.7mm whereas the District of Leh in Ladakh (UT) received the lowest rainfall of 63.8mm, which resembles with the normal pattern of rainfall in India. ***Mango showers** are pre-monsoon showers in Kerala and Karnataka. They are known as such because they help in the ripening of the mangoes.

1. India gets maximum rainfall mainly from –
 (a) North-East Monsoon (b) Retreating Monsoon
 (c) South-West Monsoon (d) Convectional rainfall

48th to 52nd B.P.S.C. (Pre) 2008

Ans. (c)

During April and May, when the Sun shines between the equator and the Tropic of Cancer (Northern Hemisphere), the large land mass in the north of Indian Ocean gets intensely heated. This causes the formation of intense low pressure in the northwestern part of the subcontinent followed by the northward shift in the position of Inter Tropical Convergence Zone (ITCZ). The south-west monsoon may thus, be seen as the continuation of the southeast trades deflected towards the Indian subcontinent after crossing the equator. The Southwest monsoon enters India in two rain-bearing systems. The first originates in the Bay of Bengal causing rainfall over the plains of north India. second is the Arabian Sea current of the south-west monsoon which brings rain to the west coast of India. According to data provided by India Meteorological Department for the year 2022, The south-west monsoon supplies almost the entire rainfall.

2. The state which receives maximum rainfall due to North-East monsoon is :

- (a) Assam (b) West Bengal
 (c) Tamil Nadu (d) Odisha

U.P.P.C.S. (Pre) 2009

Ans. (c)

Tamil Nadu, located in south-east peninsular India, receives the major part of its annual rainfall during the North-East monsoon season. Tamil Nadu receives about 40-50% of its annual rainfall during North-East monsoon.

3. Which amongst the following States gets the highest average annual rainfall?

- (a) Arunachal Pradesh (b) Sikkim
 (c) Kerala (d) Jammu and Kashmir

U.P.P.C.S. (Mains) 2016

Ans. (b)

When the question was asked option (b) was correct, but according to data provided by India Meteorological Department for the year 2022.

State/UTs	Average Annual Rainfall (mm)
Arunachal Pradesh	2810.6
Sikkim	3300.7
Kerala	2897.1
Jammu & Kashmir	1040.4
Notably, the highest average annual rainfall occurs in Meghalaya (4216.1 mm).	

4. Which among the following pairs of places have most marked differences in total annual rainfalls even though located approximately along the same latitude?

- (a) Bengaluru and Chennai
 (b) Mumbai and Vishakhapatnam
 (c) Ajmer and Shillong
 (d) Nagpur and Kolkata.

I.A.S. (Pre) 1993

Ans. (c)

Among the given pairs Ajmer (26° 27' N) and Shillong are located along the same latitude. But according to Rainfall Statistics of India, 2022, Ajmer receives 623.5 mm per annum, whereas in Shillong (East Khasi Hills District), rainfall is 8084.7 mm per annum.

5. The average annual temperature of a meteorological station is 26°C, its average annual rainfall is 63 cm and the annual range temperature is 9°C. The station in question is:

- (a) Allahabad (b) Chennai
 (c) Cherrapunji (d) Kolkata

I.A.S. (Pre) 2002

Ans. (b)

According to India Meteorological Department data, the average temperature of Chennai is 24°C in winter and 30°C in summer. The average rainfall is 60-70 cm (In 2022-140 cm.). The average temperature of Tamil Nadu for the whole year is 26°C. Its annual range of temperature is 8 to 9°C. So the correct answer is option (b).

6. Which one of the following is the correct sequence of the given Indian cities in the decreasing order of their normal annual rainfall?

- (a) Kochi-Kolkata-Delhi-Patna
 (b) Kolkata-Kochi-Patna-Delhi
 (c) Kochi-Kolkata-Patna-Delhi
 (d) Kolkata-Kochi-Delhi-Patna

I.A.S. (Pre) 2005

Ans. (c)

When the question was asked option (c) was correct. According to data of Rainfall Statistics, the correct sequence of the annual rainfall of the given cities in the year 2022 in descending order was –

Place	Rainfall (mm)
Kochi (Ernakulum)	- 3415.8
Kolkata	- 1325.1
Patna	- 736.2
Delhi	- 675.9

It is notable that the highest rainfall in India occurs at Mawsynram, Meghalaya.

7. **Mango shower is –**

- (a) Shower of mangoes
- (b) Dropping of mangoes
- (c) Rainfall in March-April in Bihar and Bengal
- (d) Crop of mango

43rd B.P.S.C. (Pre) 1994

Ans. (d)

Mango shower is the name of pre-monsoon showers in Karnataka and Kerala that help in the ripening of mangoes. It is also known as April rains or summer showers. The reason for the mango showers is the thunderstorms over the Bay of Bengal. The shower prevents mangoes from dropping prematurely.

8. **Match List-I with List-II and choose the correct answer using the codes given below in the lists:**

List-I (Famous Local Storm)	List-II (Place/Area)
A. Mango shower	1. Punjab to Bihar
B. Blossom shower	2. Kerala
C. Norwesters	3. Karnataka Coastal Areas
D. Loo	4. West Bengal

Codes :

A	B	C	D
(a) 3	2	1	4
(b) 4	1	3	2
(c) 3	2	4	1
(d) 1	3	2	4

UPPSC (Pre) 2024

Ans. (c)

The correct matching is-

<u>Famous Local Storm</u>	<u>Place/Area</u>
Mango shower	Karnataka Coastal Areas
Blossom shower	Kerala
Norwesters	West Bengal
Loo	Punjab to Bihar

- **Mango shower** - The pre monsoon stormy rainfall that occurs in Kerala and coastal Karnataka at the end of the summer season is commonly known as Mango shower as it helps in the ripening of Mangoes.
- **Blossom shower** - This rainfall in Kerala and the nearby coffee producing regions help coffee flowers to bloom.
- **Norwesters** - The violent and destructive stormy winds accompanied by rainfall that occur in Assam and West Bengal during the month of Baisakh; in the evening are called kal-Baisakhi. These rain bearing winds are beneficial for tea, jute and rice crops. In Assam, they are known as bardoli chheerha.
- **Loo** - These are dry and hot local winds & that blow across the northern great plains from Punjab to Bihar.

9. **Mango showers are common phenomena in which States of India?**

- (a) Uttar Pradesh and Madhya Pradesh
- (b) Madhya Pradesh and Karnataka
- (c) Uttar Pradesh and Tamil Nadu
- (d) Kerala and Karnataka

Jharkhand P.C.S. (Pre) 2023

Ans. (d)

See the explanation of the above.

10. **Towards the end of summer, there are pre-monsoon showers which are a common phenomenon in Kerala and coastal areas of Karnataka. Locally, they are known as :**

- (a) Mango showers
- (b) Blossom showers
- (c) Norwesters
- (d) More than one of the above
- (e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (a)

Mango showers are the name of the pre-monsoon showers in the southern states of India, specifically in the coastal areas of Karnataka and Kerala. This rain helps in the ripening of mangoes. Significantly this type of rain is also known as April rains or summer showers. It is worth mentioning that the showers prevent the mangoes from dropping prematurely.

11. **Among the following places which one gets the least rainfall?**

- (a) Leh
- (b) Bikaner

(c) Jaisalmer

(d) Cherrapunji

U.P.P.C.S. (Pre) 2008

Ans. (a)

According to the Rainfall Statistics of India, 2022 Leh receives the least rainfall in India. The annual rainfall for the above towns is as follows :

Leh	- 63.8 mm (Lowest rainfall in India)
Bikaner	- 474.0 mm
Jaisalmer	- 372.9 mm (Lowest rainfall in Rajasthan)
Cherrapunji	- 11430 mm (According to the Ministry of Jal Shakti).

12. Which one of the following places has the lowest annual rainfall ?

(a) Bellary

(b) Jaisalmer

(c) Jodhpur

(d) Leh

U.P.P.C.S. (Mains) 2010

U.P.P.C.S. (Pre) 2008, 2009

Ans. (d)

See the explanation of above question.

13. Cherrapunji is located in –

(a) Assam

(b) Manipur

(c) Meghalaya

(d) Mizoram

M.P.P.C.S. (Pre) 2005

41st B.P.S.C. (Pre) 1996

Ans. (c)

Cherrapunji (Now- Sohra) is a sub-divisional town in the East Khasi Hills district of Meghalaya. It has held the record for highest rainfall after Mawsynram.

14. The highest rainfall in India occurs in -

(a) Western Ghats, Himalayan Region and Meghalaya

(b) Madhya Pradesh and Bihar

(c) Uttar Pradesh, Haryana and Punjab

(d) Andhra Pradesh and Vidarbha

M.P.P.C.S. (Pre) 2010

Ans. (a)

The highest rainfall in India occurs in the Western Ghats, Eastern Himalayan region and Meghalaya region. These regions receive more than 250cm of average annual rainfall.

15. India has an abundant volume of rainfall but is still regarded as a thirsty land. This is due to :

(a) Rapid runoff of water

(b) Quick evaporation of rainwater

(c) Concentration of rainfall during a few months only

(d) All the above

U.P.P.C.S. (Pre) 1994

Ans. (d)

In India, almost the entire rainfall occurs due to the southwest monsoon from June to September. Water from rainfall due to this monsoon either flows away or evaporates quickly. In this way, India has to face a scarcity of rainfall in the remaining 8 months. That is why besides having 1257 mm of actual annual rainfall, it is regarded as a thirsty land.

16. Which of the following Water Management Strategies in India is cost-effective ?

(a) Improvement of the flusti system

(b) Treatment and use of wastewater

(c) Rainwater Harvesting

(d) Recycling and Re-use of Water

U.P.P.C.S. (Mains) 2011

Ans. (c)

Rainwater harvesting is the water management strategy that is cost-effective in India. It can be done by building embankments and using tanks. It also increases the water level. Tamil Nadu is the leading state in rainwater harvesting.

17. Which of the following areas have an average of 200mm rainfall in India ?

(a) Kerala, Tamil Nadu, Karnataka

(b) Jammu and Kashmir

(c) West Bengal, Orissa, Bihar

(d) Assam, Manipur, Tripura

U.P.P.C.S. (Pre) 1994

Ans. (b)

During the question period, the northern part of the Zaskar (Zaskar) range in Jammu and Kashmir (Now- Ladakh) and Western part of Rajasthan receive an average annual rainfall of 200mm (20cm). Hence, option (b) can be considered as the correct answer. However, from the perspective of the overall state none of the above will be, correct answer.

18. By which monsoon does Jharkhand receive rainfall?

(a) North East Monsoon

(b) South West Monsoon

(c) North West Mansoon

(d) None of these

Jharkhand P.C.S. (Pre) 2011

Ans. (b)

Jharkhand receives its rainfall during south-west monsoon i.e., June to September.